

API SECURITY RISK ANALYSIS REPORT

Future Interns — Cyber Security Internship

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Tools Used:
Postman
Browser DevTools
Public Test APIs

INTRODUCTION

Modern applications rely heavily on APIs for communication between applications, mobile platforms, SaaS systems, and cloud services.

Insecure APIs may expose sensitive information, increase attack surfaces, and allow unauthorized access to systems and data.

This project focuses on identifying API security risks using ethical and non-intrusive analysis techniques.

OBJECTIVE

- Analyze public API endpoints
- Identify common API security risks
- Review authentication practices
- Assess data exposure risks
- Suggest remediation strategies
- Improve API security awareness

TOOLS USED

- Postman
- Browser DevTools
- JSONPlaceholder Test API
- Canva / Documentation Tools

Purpose:

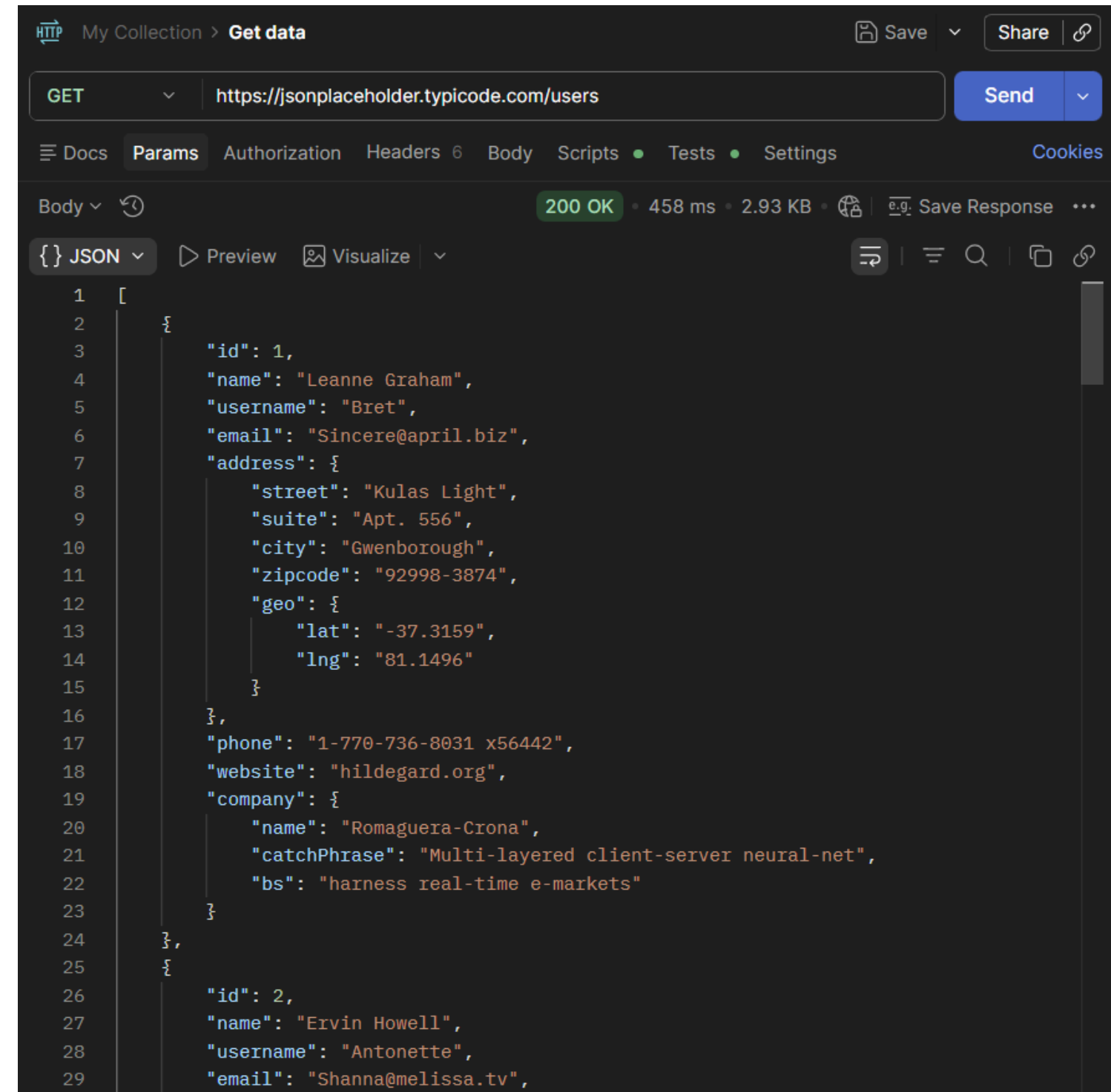
To inspect API endpoints and analyze security risks ethically and safely.

TESTED API ENDPOINTS

- 1./users
- 2./posts
- 3./comments

Observations:

- Public accessibility
- JSON response exposure
- No visible authentication requirements



```
HTTP My Collection > Get data Save Share
GET https://jsonplaceholder.typicode.com/users Send
Docs Params Authorization Headers 6 Body Scripts Tests Settings Cookies
Body 200 OK 458 ms 2.93 KB Save Response
JSON Preview Visualize
1 [
2   {
3     "id": 1,
4     "name": "Leanne Graham",
5     "username": "Bret",
6     "email": "Sincere@april.biz",
7     "address": {
8       "street": "Kulas Light",
9       "suite": "Apt. 556",
10      "city": "Gwenborough",
11      "zipcode": "92998-3874",
12      "geo": {
13        "lat": "-37.3159",
14        "lng": "81.1496"
15      }
16    },
17    "phone": "1-770-736-8031 x56442",
18    "website": "hildegard.org",
19    "company": {
20      "name": "Romaguera-Crona",
21      "catchPhrase": "Multi-layered client-server neural-net",
22      "bs": "harness real-time e-markets"
23    }
24  },
25  {
26    "id": 2,
27    "name": "Ervin Howell",
28    "username": "Antonette",
29    "email": "Shanna@melissa.tv",
```

GET

https://jsonplaceholder.typicode.com/posts

Send

Docs

Params

Authorization

Headers 6

Body

Scripts

Tests

Settings

Cookies

Body

200 OK • 588 ms • 7.98 KB • Save Response

JSON

Preview

Visualize

```
1  [
2    {
3      "userId": 1,
4      "id": 1,
5      "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
6      "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet architecto"
7    },
8    {
9      "userId": 1,
10     "id": 2,
11     "title": "qui est esse",
12     "body": "est rerum tempore vitae\nsequi sint nihil reprehenderit dolor beatae ea dolores neque\nfugiat blanditiis voluptate porro vel nihil molestiae ut reiciendis\nqui aperiam non debitis possimus qui neque nisi nulla"
13   },
14   {
15     "userId": 1,
16     "id": 3,
17     "title": "ea molestias quasi exercitationem repellat qui ipsa sit aut",
18     "body": "et iusto sed quo iure\nvoluptatem occaecati omnis eligendi ad\nvoluptatem doloribus vel accusantium quis pariatur\nmolestiae porro eius odio et labore et velit aut"
19   },
20   {
21     "userId": 1,
```

GET

https://jsonplaceholder.typicode.com/comments

Send

Docs

Params

Authorization

Headers 6

Body

Scripts

Tests

Settings

Cookies

Body

200 OK • 658 ms • 40.77 KB • Save Response

JSON

Preview

Visualize

```
1  [
2    {
3      "postId": 1,
4      "id": 1,
5      "name": "id labore ex et quam laborum",
6      "email": "Eliseo@gardner.biz",
7      "body": "laudantium enim quasi est quidem magnam voluptate ipsam eos\ntempora quo necessitatibus\ndolor quam autem quasi\nreiciendis et nam sapiente accusantium"
8    },
9    {
10     "postId": 1,
11     "id": 2,
12     "name": "quo vero reiciendis velit similique earum",
13     "email": "Jayne_Kuhic@sydney.com",
14     "body": "est natus enim nihil est dolore omnis voluptatem numquam\net omnis occaecati quod ullam at\nvoluptatem error expedita pariatur\nnihil sint nostrum voluptatem reiciendis et"
15   },
16   {
17     "postId": 1,
18     "id": 3,
19     "name": "odio adipisci rerum aut animi",
20     "email": "Nikita@garfield.biz",
21     "body": "quia molestiae reprehenderit quasi aspernatur\naut expedita occaecati aliquam eveniet laudantium\nnomnis quibusdam delectus saepe quia accusamus maiores nam est\ncum et ducimus et vero voluptates excepturi deleniti ratione"
22   },
```


ENDPOINT ANALYSIS

/users

- User-related data publicly accessible

/posts

- Application data accessible without authentication

/comments

- Public exposure of comment-related data

Risk Classification:

MEDIUM

IDENTIFIED API SECURITY RISKS

- Missing authentication controls
- Excessive data exposure
- Public endpoint accessibility
- Lack of visible rate limiting

Potential Impact:

Unauthorized access, information disclosure, and abuse risks.

BUSINESS IMPACT

Insecure APIs may result in:

- Exposure of sensitive information
- Increased attack surface
- Unauthorized access risks
- Compliance and reputation concerns

API security is critical for modern SaaS and cloud applications.

REMEDIATION RECOMMENDATIONS

- Implement authentication mechanisms
- Apply authorization controls
- Reduce unnecessary data exposure
- Enable API rate limiting
- Monitor API traffic
- Validate user inputs securely

CONCLUSION

The API security assessment identified common API security concerns related to authentication, public accessibility, and excessive data exposure.

The project demonstrates the importance of secure API design and strong security controls in modern applications.

The assessment was conducted ethically using public test APIs and non-intrusive analysis methods.

THANK YOU

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